AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 28 · STUDENT LESSON

Transfer learning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Freeze the extractor. Fit the head. Check the evidence.

Trace a supplied representation and a training-fitted threshold, then compare both methods on identical validation rows.

Keep one habit: Parameter map → complete training table → executable audit → validation → separate picture → collision → independent extractor.

MAIN LESSON · SECTION 1

Reuse a representation, fit a new decision rule

> Track exactly which parameters stay frozen and which are fitted.

Learning goals

I can trace a reused **feature extractor** and a newly fitted **output head**.

I can fit a small target-task head using **training** data and compare a baseline.

I can identify when a frozen representation cannot support a new task.

Retrieve representation and comparison

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and response

- If two different records become the same feature vector, can a deterministic **output head** receiving only that vector distinguish them?

> **My starting explanation**

End goal

By the end, I can explain **transfer learning** using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Trace a reused **feature extractor** and a newly fitted **output head**

Fit a small target-task head using **training** data and compare a baseline

Identify when a frozen representation cannot support a new task

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Separate the frozen extractor from the fitted head

> Our supplied extractor demonstrates reuse mechanics; it has no source-training history.

Transfer learning and the transparent stand-in

A source model may learn useful features before we have many labels for a new target task. **Transfer learning** reuses those learned features or parameters. One approach freezes the **feature extractor** and fits a new **output head**.

Fine tuning instead updates some or all of the reused parameters, usually together with the head.

Use a transparent stand-in for a source-trained extractor: f(x) = ReLU(x1 + x2), with supplied weights [1, 1] and bias 0. ReLU keeps a positive value and replaces a negative value with zero.

Here x1 and x2 are nonnegative fictional event counts, so f(x) is their sum. We did not train this extractor on a source dataset; this exercise demonstrates the mechanics of reuse, not measured transfer benefits.

Follow input → feature → prediction

Part	Operation	During this exercise
Input	[count of type 1, count of type 2]	Keep order and units

Part	Operation	During this exercise
Extractor	h = max(0, x1 + x2)	Freeze weights and bias
New head	Predict 1 if h >= threshold	Fit the threshold on target training data

Interpret the feature and record missing facts

- The target is a high-activity label. The extractor returns a single number h, not a class label or probability. A new head converts h into the target **prediction**.

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Fit the threshold using four training records

> Compute each feature once, then compare the three candidate heads.

Complete training inputs and fitting rule

Use target **training** records T1 [1, 1] with label 0, T2 [1, 3] with label 0, T3 [4, 2] with label 1 and T4 [5, 3] with label 1. Their extracted values are 2, 4, 6 and 8.

Fit the head by trying thresholds 3, 5 and 7 and counting **training** mistakes. **Fitting** can use a small search; it need not use gradients.

Trace every training feature and prediction

Training record · Input counts · Label · Frozen h · Prediction at threshold 5

> **Training record:** T1
> **Input counts:** [1, 1]
> **Label:** 0
> **Frozen h:** 2
> **Prediction at threshold 5:** 0

> **Training record:** T2
> **Input counts:** [1, 3]
> **Label:** 0
> **Frozen h:** 4
> **Prediction at threshold 5:** 0

> **Training record:** T3
> **Input counts:** [4, 2]
> **Label:** 1
> **Frozen h:** 6
> **Prediction at threshold 5:** 1

> **Training record:** T4
> **Input counts:** [5, 3]
> **Label:** 1
> **Frozen h:** 8
> **Prediction at threshold 5:** 1

> **Training record:** Candidate threshold 3
> **Input counts:** Training mistakes

> **Label:** 1
>
> **Frozen h:** Threshold 5: 0
>
> **Prediction at threshold 5:** Threshold 7: 1

Training tie rule and baseline rule

The tie rule prefers the smaller threshold if **training** mistake counts match; there is no tie here. The extractor remains unchanged. The code assumes finite numeric counts and valid binary labels.

A constant baseline predicts the most frequent **training** label; with two of each, our declared tie rule chooses 0.

Run the complete fit and validation audit

```
"""Supplied frozen extractor; fit only the target threshold on training rows."""
from math import isfinite
train = [('T1', [1, 1], 0), ('T2', [1, 3], 0),
         ('T3', [4, 2], 1), ('T4', [5, 3], 1)]
validation = [('V1', [2, 1], 0), ('V2', [3, 4], 1)]
def feature(x):
    if len(x) != 2 or any(not isfinite(v) or v < 0 for v in x):
        raise ValueError('Need two finite nonnegative counts')
    return max(0, x[0] + x[1])
def predict(x, threshold):
    return int(feature(x) >= threshold)
def errors(threshold):
    return sum(predict(x, threshold) != y for _, x, y in train)
candidates = [3, 5, 7]
best = min(candidates, key=lambda t: (errors(t), t))
counts = {y: sum(label == y for _, _, label in train) for y in (0, 1)}
baseline = min(counts, key=lambda y: (-counts[y], y))
ledger = [(name, feature(x), y, predict(x, best), baseline)
          for name, x, y in validation]
head_correct = sum(pred == y for _, _, y, pred, _ in ledger)
baseline_correct = sum(base == y for _, _, y, _, base in ledger)
assert [errors(t) for t in candidates] == [1, 0, 1]
assert (best, baseline, head_correct, baseline_correct) == (5, 0, 2, 1)
assert predict([2, 3], best) == 1  # equality belongs to class 1
print('Threshold mistakes:', [(t, errors(t)) for t in candidates])
print('Training label counts:', counts, 'baseline:', baseline)
```

```
print('Validation: name, h, label, head, baseline', ledger)
print('Correct on same two rows:', head_correct, baseline_correct)
# Different target: first count exceeds second. Both original features collide.
u, v = [1, 3], [3, 1]
assert feature(u) == feature(v) == 4
# This repair is hand supplied, not trained or independently evaluated.
def supplied_repair(x):
    return int(max(0, x[0] - x[1]) >= 1)
assert (supplied_repair(u), supplied_repair(v)) == (0, 1)
print('Collision h:', feature(u), feature(v), 'supplied repair:',
      supplied_repair(u), supplied_repair(v))
```

Explain each visual step

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Reference: original compact fitting code

```
train = [([1, 1], 0), ([1, 3], 0),
         ([4, 2], 1), ([5, 3], 1)]

def feature(x):
    if len(x) != 2 or any(v < 0 for v in x):
        raise ValueError("Need two nonnegative counts")
    return max(0, x[0] + x[1])

def predict(x, threshold):
    return int(feature(x) >= threshold)

def mistakes(threshold):
    return sum(predict(x, threshold) != y
               for x, y in train)

candidates = [3, 5, 7]
best = min(candidates, key=lambda t: (mistakes(t), t))
assert best == 5
print([mistakes(t) for t in candidates])  # [1, 0, 1]
```

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change one input while keeping fitted parameters fixed

> Predict first; inference does not refit the head.

Current task and response

- For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

  Explain which data fitted the head.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed picture

- Change: Query changes to [2, 1].

▼ Check after predicting

Prediction: Its h becomes 4 and the fitted head predicts class 0.

Reason: Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Validate the fitted head, then interpret a separate picture

> Use the same validation rows for the head and training-majority baseline.

Main validation results and evidence limits

Freeze the fitted head before validation

VALIDATION INPUT	FROZEN FEATURE	HEAD AT THRESHOLD 5
V1 [2, 1], label 0	h = 3	Predict 0, correct
V2 [3, 4], label 1	h = 7	Predict 1, correct

The fitted head is correct on 2 of 2 validation records; the constant class 0 baseline is correct on 1 of 2. This tiny constructed comparison is not evidence of general transfer performance.

Use the same validation records for both methods. These records were not used to fit the threshold. If validation guides later choices, reserve separate untouched final evaluation data.

We have not compared with a full model trained from scratch or measured **training** time, so this result cannot establish either advantage.

Try the equality case before checking

A new input [2, 3] uses the fitted threshold 5. Give h and the predicted class, including the equality rule. Which parameters changed during fitting?

Separate pictured extractor and complete givens

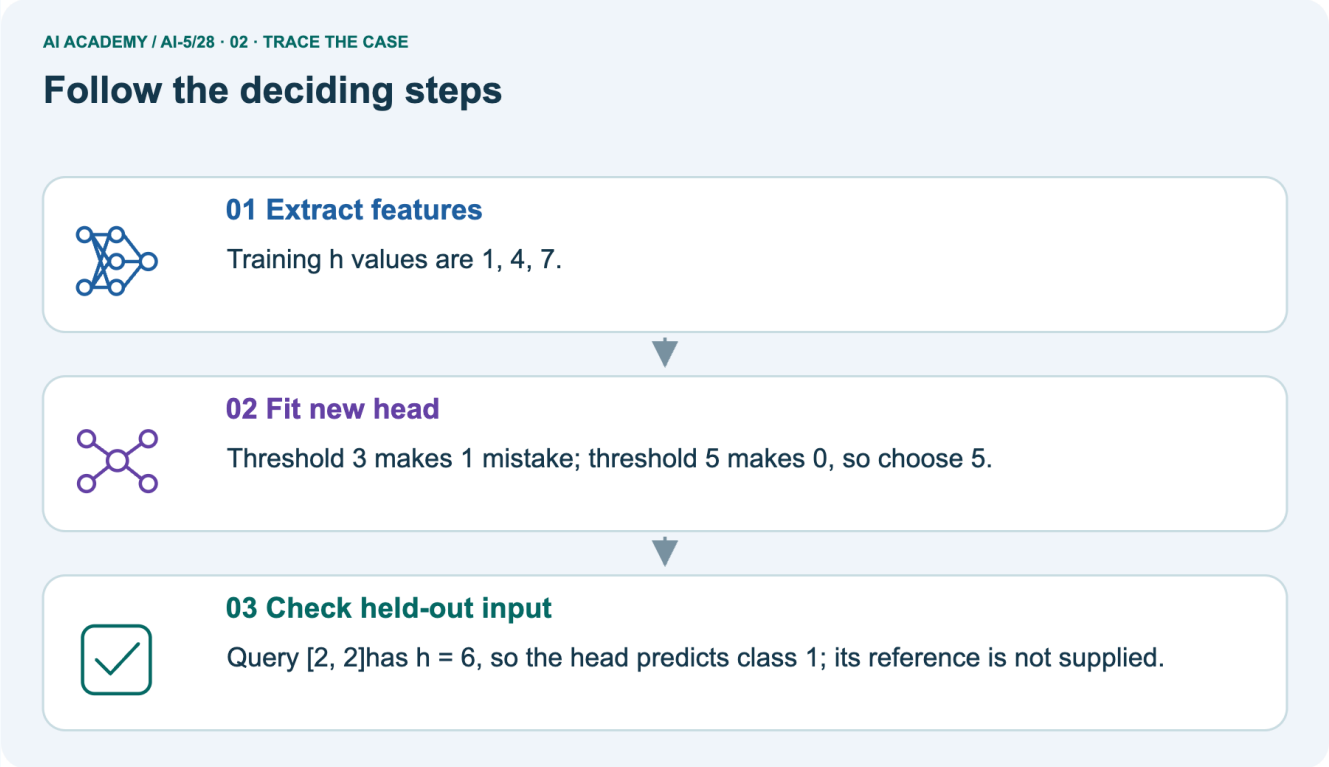
Transfer a fixed **feature extractor** while identifying what is actually refitted.

Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1. Class 1 when h >=threshold; candidates 3 or5.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep two extractors separate. The main example freezes h = x1 + x2 and fits threshold 5. Its validation inputs have supplied labels, so correctness can be counted. The picture freezes h = left + 2 × right: training features are 1, 4, 7, and threshold 5 makes zero training mistakes. Its query [2, 2] gives h = 6 and class 1, but has no supplied reference label. A prediction alone cannot be scored as correct.

Trace the pictured fitting and inference



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Extract features	Training h values are 1, 4, 7.
2. Fit new head	Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.
3. Check held-out input	Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

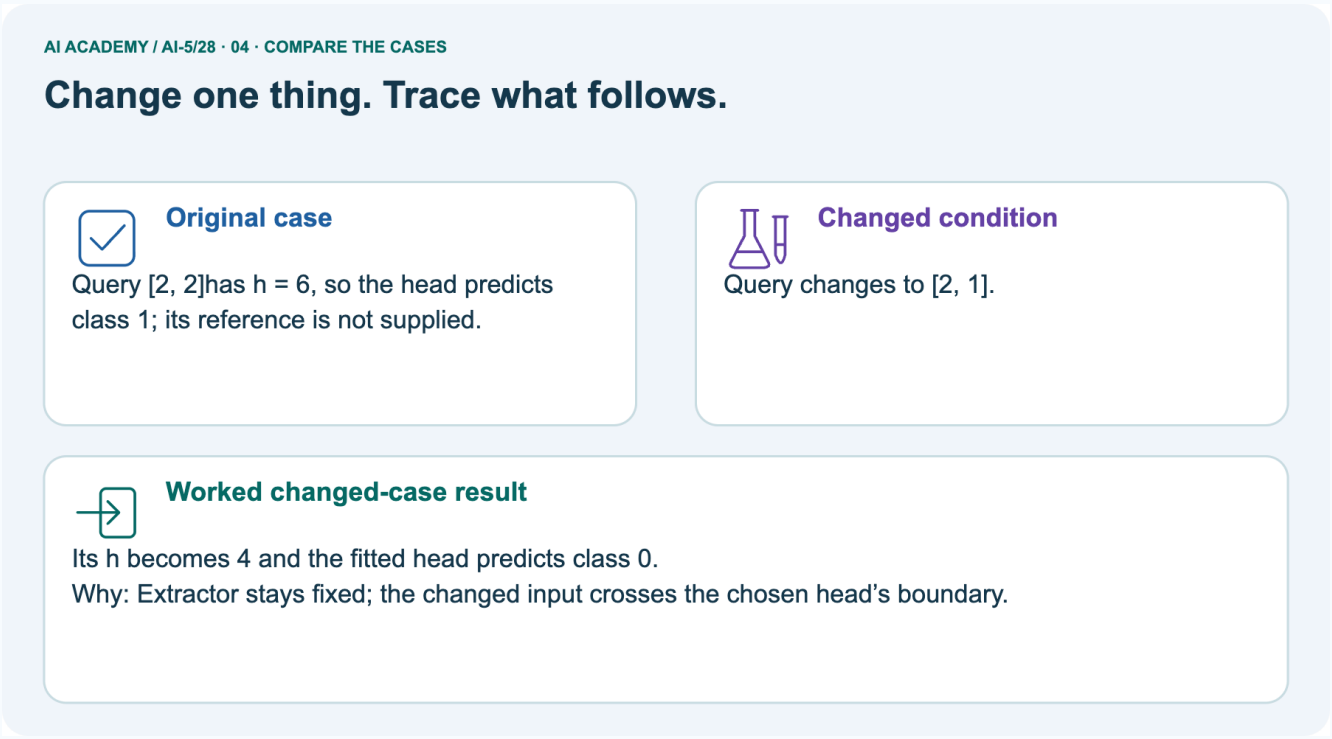
Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Predict the changed query

- Change one thing: Query changes to [2, 1]. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what was fitted and what remains unknown

- A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

- Read the labelled picture: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

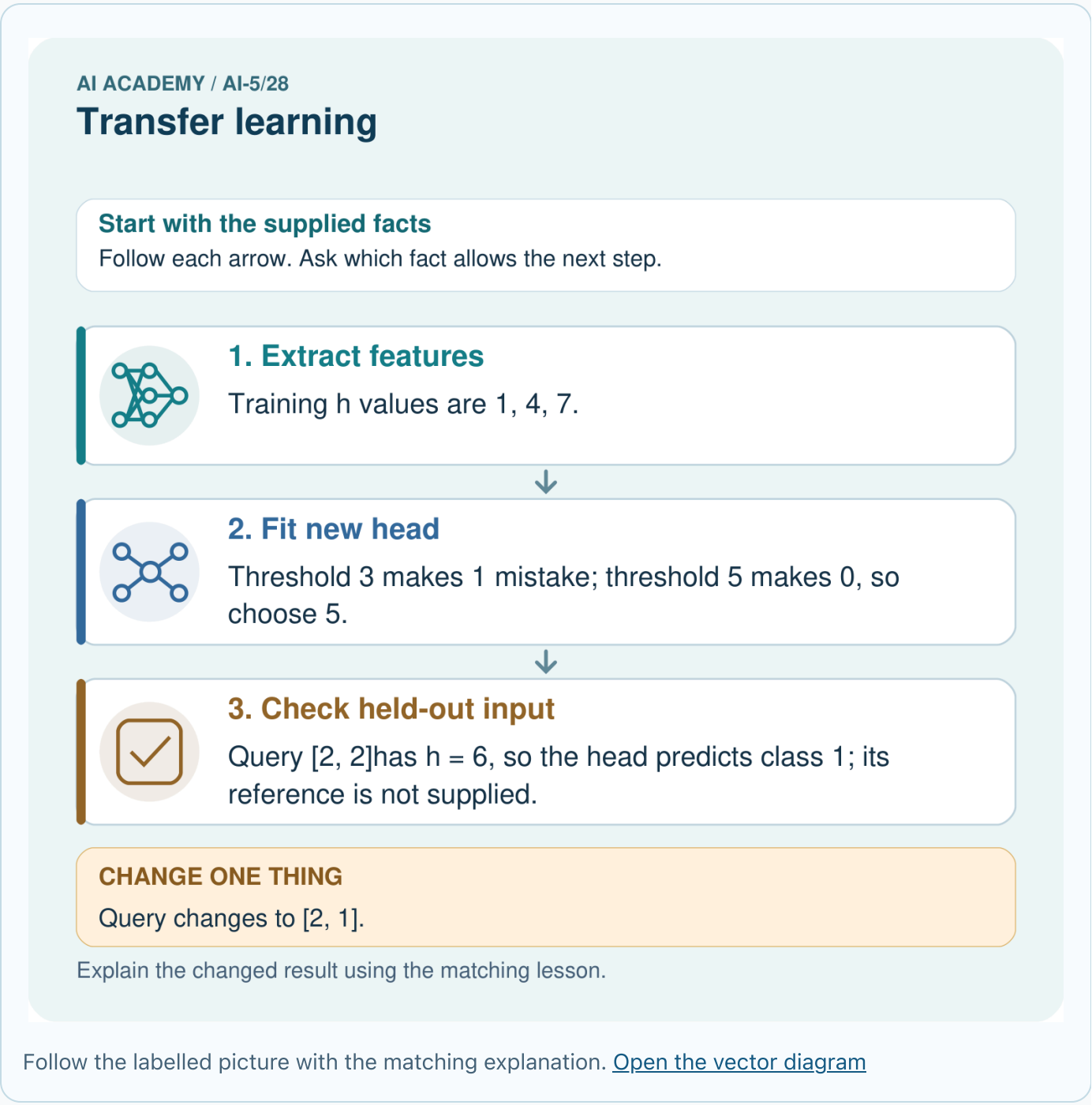
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Transfer learning

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Check whether the frozen representation keeps the needed distinction

> A better head cannot recover information that the feature already discarded.

Reason about a changed target

Now change the target: predict 1 when the first count is greater than the second, otherwise 0. U [1, 3] has label 0 and V [3, 1] has label 1, but both become h = 4.

Every fixed deterministic head receiving only this h must give both the same **prediction**. At least one of the two will be wrong, regardless of how many extra head parameters we add.

Fine tuning could change the extractor to retain relevant distinctions; it does not guarantee success. To see a possible change, hand-set the weights to [1, -1], keep bias 0 and use threshold 1.

U then gives ReLU(-2) = 0 and predicts 0; V gives ReLU(2) = 2 and predicts 1. This hand-chosen repair is not a trained result or an independent evaluation.

Keeping the original **input** alongside h is another possible representation change. Any such change needs a new **training** and evaluation plan.

Normalising the **input**s also changes the pipeline; last week showed why that can remove useful magnitude information.

Complete collision and explicitly supplied repair

New target: first count exceeds second	Supplied input	Correct label	Original frozen h
U	[1, 3]	0	4
V	[3, 1]	1	4

New target: first count exceeds second	Supplied input	Correct label	Original frozen h
Collision	Same h, conflicting labels	Any deterministic h-only head gives the same output	Cannot classify both correctly
Hand-set repair	h = ReLU(x1 – x2), threshold 1	U → 0; V → 1	Representation changed; no training run claimed

Try the misconception check

- For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1.

  Does this show that **fine tuning** has been trained and independently validated?

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise fitting, scoring and diagnosing separately

> Use target training labels to fit; use validation labels to compare.

Current practice and response

- For **input** [4, 2], compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

A new input [2, 3] uses the fitted threshold 5. Give h and the predicted class, including the equality rule. Which parameters changed during fitting?

Explain your prediction to a partner before comparing results.

Supported W2 record

Use workbook W2 for the connected example "**Transfer learning**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Misconception check

- Use the worked case for **Transfer learning**. Proposed explanation: “Frozen means the whole model cannot learn”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture reasoning

- Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the new independent extractor

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Fit and compare

Check your reasoning

Explain the picture

Keep a complete fit-and-compare evidence record

> The executable audit is in Section 3; record actual runs separately from paper traces.

Current investigation and original procedure

For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

Explain which data fitted the head.

```
train = [([1, 1], 0), ([1, 3], 0),
         ([4, 2], 1), ([5, 3], 1)]

def feature(x):
    if len(x) != 2 or any(v < 0 for v in x):
        raise ValueError("Need two nonnegative counts")
    return max(0, x[0] + x[1])

def predict(x, threshold):
    return int(feature(x) >= threshold)

def mistakes(threshold):
    return sum(predict(x, threshold) != y
               for x, y in train)

candidates = [3, 5, 7]
best = min(candidates, key=lambda t: (mistakes(t), t))
assert best == 5
print([mistakes(t) for t in candidates])  # [1, 0, 1]
```

Expected outputs, input checks and honest limits

1. Predict threshold mistakes [1, 0, 1], then run or paper-trace the complete program in Section 3. Save the chosen threshold 5, training class counts {0:2, 1:2}, baseline class 0 and both validation rows. The fitted head scores 2/2 and the baseline 1/2 on these same constructed rows. Check equality at h = 5 and reject missing, negative or nonfinite inputs under the helper contract.

   For a new target, trace the U/V collision and label the repair as hand supplied. No source dataset, pretrained checkpoint, from-scratch training, timing or untouched final evaluation is supplied here.

Evidence ledger and retained explanation

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: fit a head on a different frozen extractor

> Use the new formula and training-only tie rule before scoring validation.

Complete independent inputs and task

A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Fit a new head independently

Attempt this without hints

A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the extractor and head as separate project artifacts

> A reusable record needs feature definition, fitting data, tie policy and evaluation scope.

Project task and artifact

Add a labelled **transfer learning evidence** sheet to Image-pattern Network Investigation. Show how you can trace a reused **feature extractor** and a newly fitted **output head**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain what was reused, fitted and actually measured

> Finish with a parameter account and a claim supported by the available evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Trace a reused **feature extractor** and a newly fitted **output head**

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Transfer learning	Reusing knowledge learned on a source task or dataset to help a target task
Feature extractor	The part of a model that maps an input to a representation
Output head	The part that maps a representation to the current task's output
Fine tuning	Training some or all previously learned parameters on the target task

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For **input** [4, 2], compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

- For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1.

  Does this show that **fine tuning** has been trained and independently validated?

- A report omits the source checkpoint and pre**process**ing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set.

  Name the missing records and propose a fair comparison with a model trained from scratch.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the independent h = 2*x1 + x2 case for the first individual attempt.

Current explanation and response

- For a real source-trained model, record the checkpoint version, source task and data provenance, permitted use, **input** order and pre**process**ing, frozen and trainable parts, target splits, baseline and resource budget.

  Check for source-data overlap with target evaluation records; a different filename is not proof of independence. If source history is unknown, report that limitation.

- Evaluate methods on the same target examples under a declared **training** budget, with choices made before final evaluation. Record errors as well as totals.

  Next week examines uncertainty and failure analysis, including confident **prediction**s that can still be wrong.

- A report omits the source checkpoint and pre**process**ing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set.

  Name the missing records and propose a fair comparison with a model trained from scratch.

A team says its reused extractor must outperform training from scratch because it scored 2 of 2 here. Give two reasons this conclusion is unsupported and propose a fair next comparison.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Freeze the fitted head before validation

VALIDATION INPUT	FROZEN FEATURE	HEAD AT THRESHOLD 5
V1 [2, 1], label 0	h = 3	Predict 0, correct
V2 [3, 4], label 1	h = 7	Predict 1, correct

The fitted head is correct on 2 of 2 validation records; the constant class 0 baseline is correct on 1 of 2. This tiny constructed comparison is not evidence of general transfer performance.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Transfer a fixed **feature extractor** while identifying what is actually refitted.

By the end, I can	How I will show it
Trace a reused feature extractor and a newly fitted output head	Explain the deciding step, show a trace or test, and state one limit.
Fit a small target-task head using training data and compare a baseline	Explain the deciding step, show a trace or test, and state one limit.
Identify when a frozen representation cannot support a new task	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Transfer a fixed **feature extractor** while identifying what is actually refitted.
- Which labelled step matches this detail? Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.
- 1. Check held-out **input**
- 2. Extract features
- 3. Fit new head
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check held-out **input** Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Changed-case question and optional hint

- Changed-case practice: Query changes to [2, 1]. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Its h becomes 4 and the fitted head predicts class 0.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Its h becomes 4 and the fitted head predicts class 0. Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

Repair and retain the record

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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<summary>Reference · Six-picture concept journey</summary>

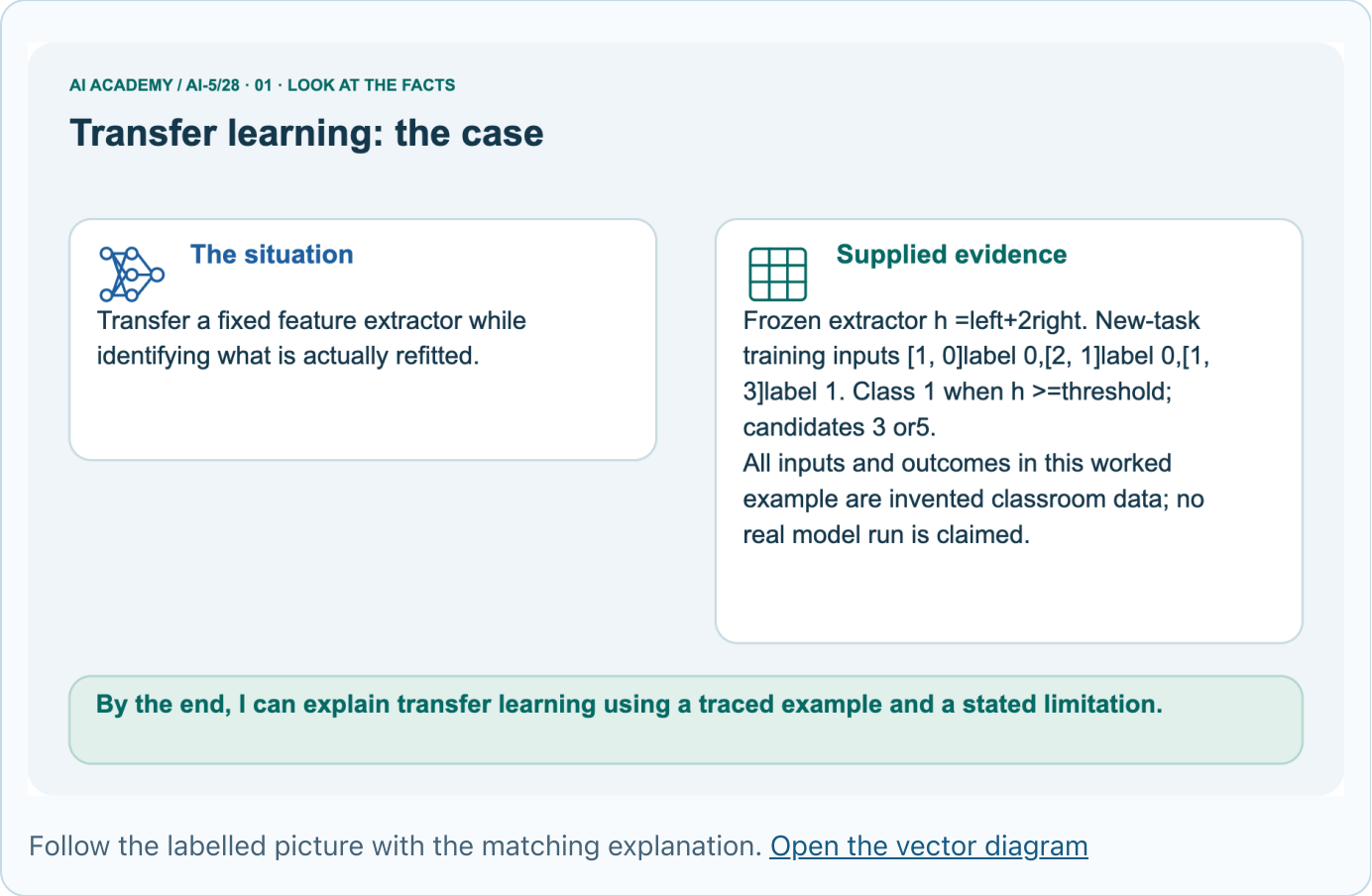
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REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Transfer learning: the case



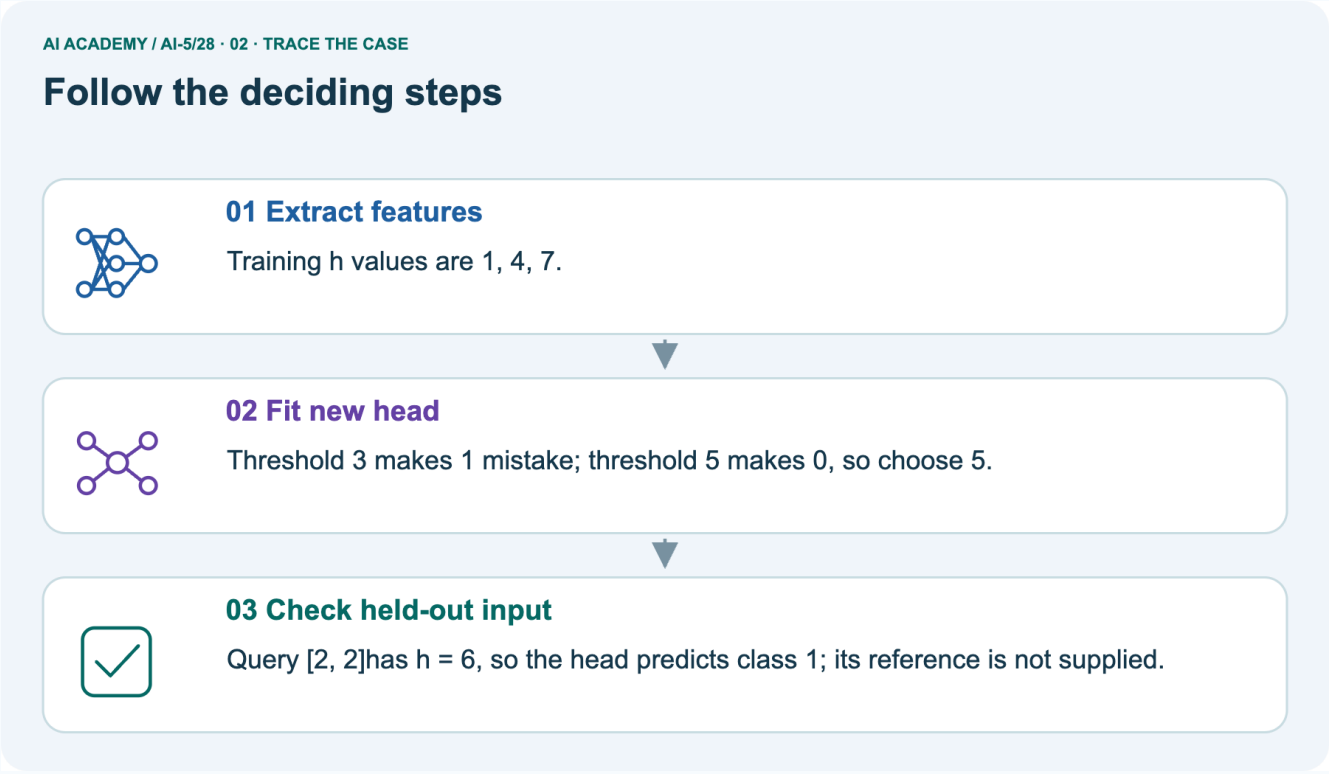
Follow the labelled picture with the matching explanation. Open the vector diagram

Transfer a fixed **feature extractor** while identifying what is actually refitted. Supplied facts: Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1.

Class 1 when h >=threshold; candidates 3 or5. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

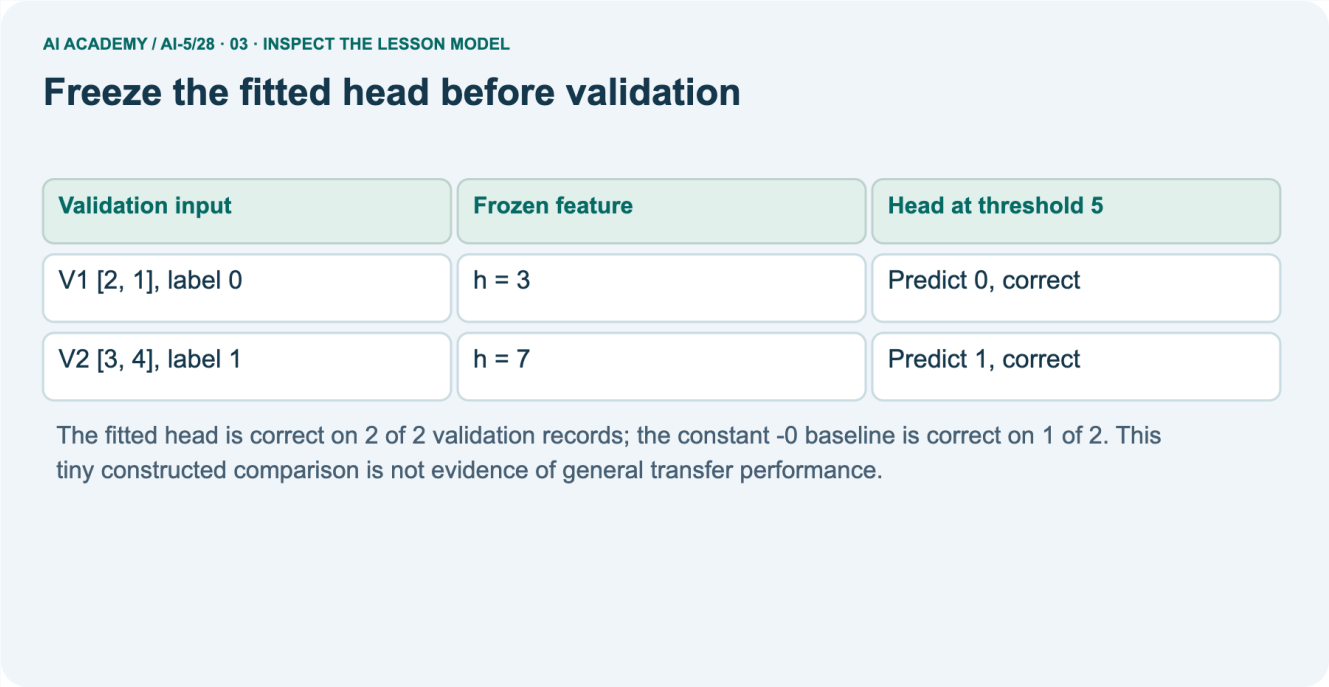


Follow the labelled picture with the matching explanation. Open the vector diagram

Extract features: **Training** h values are 1, 4, 7. → Fit new head: Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5. → Check held-out **input**: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

03. Freeze the fitted head before validation

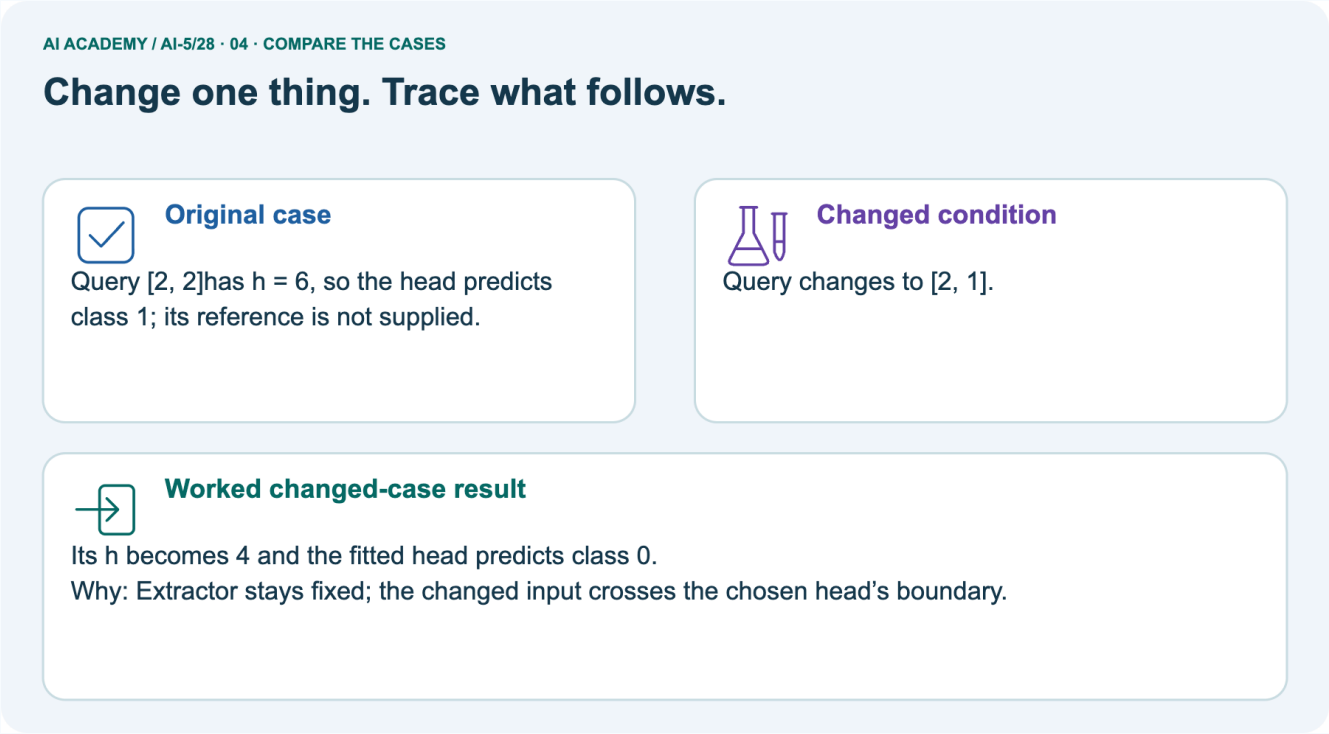


Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze the fitted head before validation. V1 [2, 1], label 0 | h = 3 | Predict 0, correct. V2 [3, 4], label 1 | h = 7 | Predict 1, correct

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



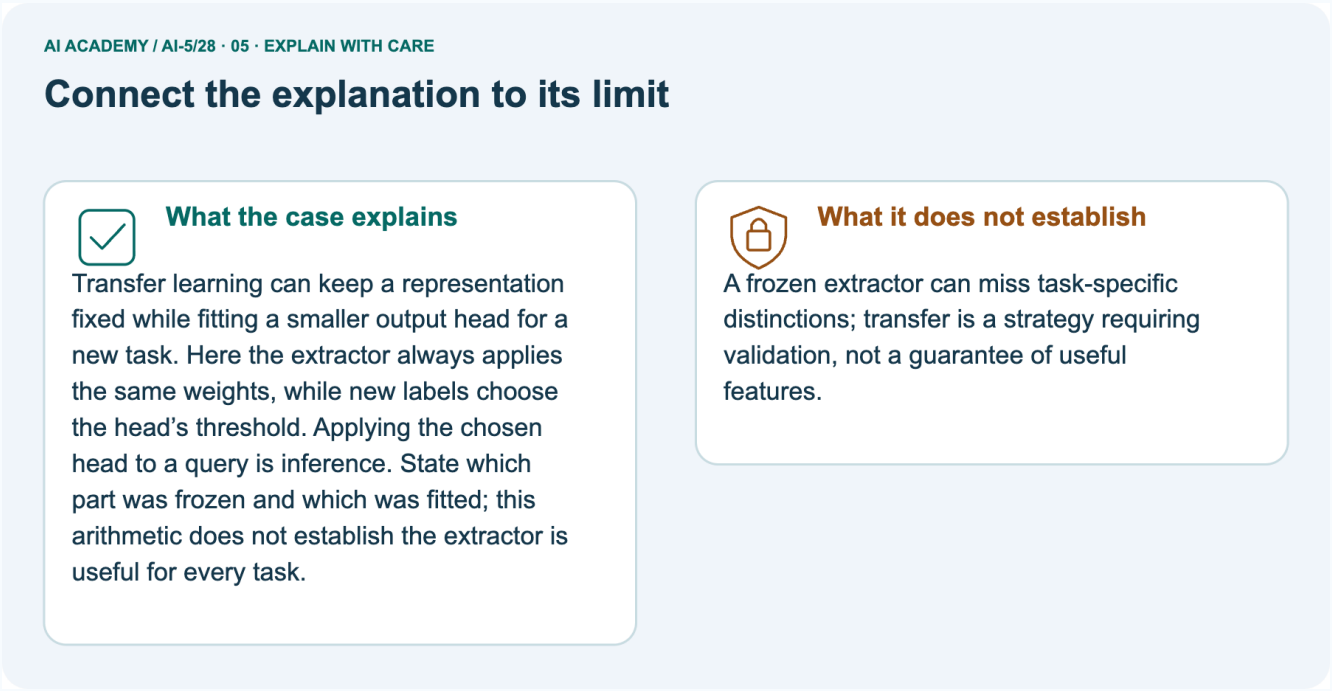
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied. Changed condition: Query changes to [2, 1]. Expected result: Its h becomes 4 and the fitted head predicts class 0.

Reason: Extractor stays fixed; the changed **input** crosses the chosen head’s boundary.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

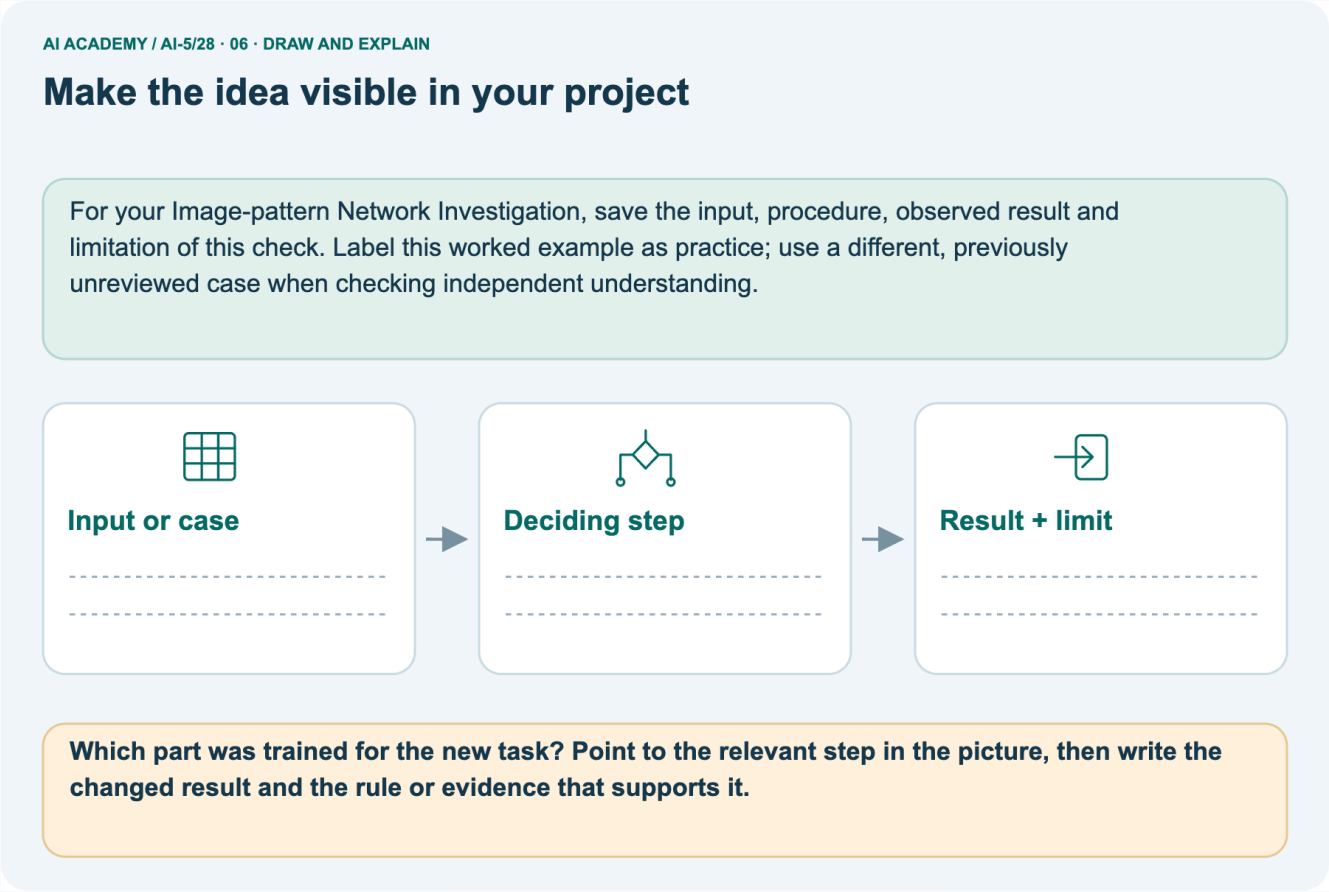
Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

Limit: A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Transfer learning: the case

02. Follow the deciding steps

03. Freeze the fitted head before validation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.